

Nguyen Thanh Long

Full-Stack Engineer

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SUMMARY

Full-stack engineer, 2.5 years, working in TypeScript on both ends of a production IoT platform: the Next.js dashboard partners use and the NestJS APIs behind it. I am currently the only engineer on the cloud side, so the same work also covers the database, the message pipeline and the deployments.

EXPERIENCE

Full-Stack Engineer at Rogo Solutions JSC Jan 2024 to present · Hanoi

A 20-person IoT startup. I am the only engineer on the cloud side, so the dashboard, the API, the message pipeline and the deploy path are all mine. The platform serves 640k users and 69k devices for 26 partner projects, at roughly 2.6M API requests a day and 15.5k concurrent MQTT connections.

- I build both ends of the partner dashboard, the Next.js pages and the NestJS endpoints behind them. The organisations area is mine end to end: list and detail pages, an invite flow that finds members by email, project assignment, and the API and data model underneath.
- One deployment serves 26 partner projects, so a lot of the work is access control. I own the authorisation rules on the server and the matching checks in the React app, so a partner is not offered an action the API would refuse. Settings that used to be global, such as the SMS provider, are now configured per project, with the setup page reduced to three questions.
- I run the pipeline that carries device state. Partner traffic used to go through one Kafka broker with no replication, so I moved it onto a replicated three-broker cluster with a separate listener for partner systems. The consumer merges repeated state updates instead of writing every one.
- There was no monitoring when I started. Prometheus, Loki and Grafana now cover the three clusters with the AWS side pulled in from CloudWatch, so an incident starts at a dashboard instead of a guess. I also own the staging to production path across four repositories.
- Most of my performance work has been finding the cause rather than writing anything clever. A device query took eleven minutes because nothing was indexed behind a \$lookup, and the plan had been to pay for a bigger database tier instead. Auth middleware was verifying a token signature on every request. A partner SDK sent everything at once on the hour. Each fix was small once the cause was clear.
- Smaller things that made the work easier for everyone: the test suite had been abandoned because running it exhausted memory, so I moved it to Jest 29 with transpile-only tests and put it back in CI; the dashboard now survives a CMS outage instead of crashing, and shows the deployed commit in a corner badge.

SELECTED PROJECTS

Educrysal (Next.js, Supabase, Tailwind, PWA) · educrysal.vercel.app
A guided crystal-growing app for children: step-by-step programs, a private photo diary, and a gallery a moderator approves. Access rules live in Postgres row level security rather than in the application, so a route cannot leak another child's diary. Installable as a PWA. Six months of work.

IoT Cloud MCP (TypeScript, NestJS, MCP SDK, Redis)
An MCP server for IoT cloud platforms, so an assistant can query and control devices through one interface rather than a separate integration per platform.

Novel Studio (NestJS, PostgreSQL with pgvector)
A long-form writing tool. The awkward part is that prompt cost grows with the story, so it keeps a canon store, a rolling summary and vector retrieval instead of resending history.

TECHNICAL SKILLS

Frontend	TypeScript, React, Next.js, Tailwind CSS, Ant Design, React Query, React Native (Expo)
Backend	Node.js, NestJS, Express, REST APIs, WebSockets
Data	PostgreSQL, MongoDB, Redis, MySQL
Infrastructure	Docker, Kubernetes, AWS (ECS, S3, Lambda), Terraform, GitHub Actions
Messaging & auth	Kafka, MQTT, Firebase Auth, Keycloak, JWT

EDUCATION & ADDITIONAL

B.Sc. in Software Engineering, FPT University 2022 to 2026
Final-year project: a real-time multiplayer game backend (ASP.NET Core, WebSockets, MySQL) using lag compensation to keep play fair across uneven network conditions.

Certifications: Project Management Principles (UC Irvine, 2025) · Software Development Lifecycle Specialization (University of Minnesota, 2024).
Coach competitive-programming students for national Informatics examinations, weekly since 2024. Freelance backend work alongside full-time employment. English: professional working proficiency.